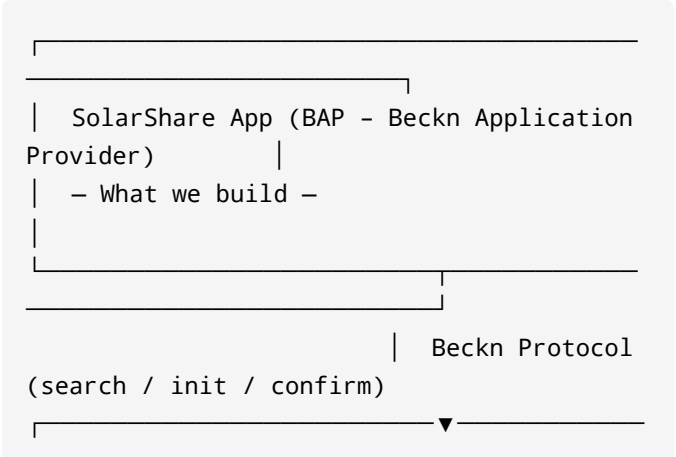
SolarShare — Technical Requirements Document (TRD)

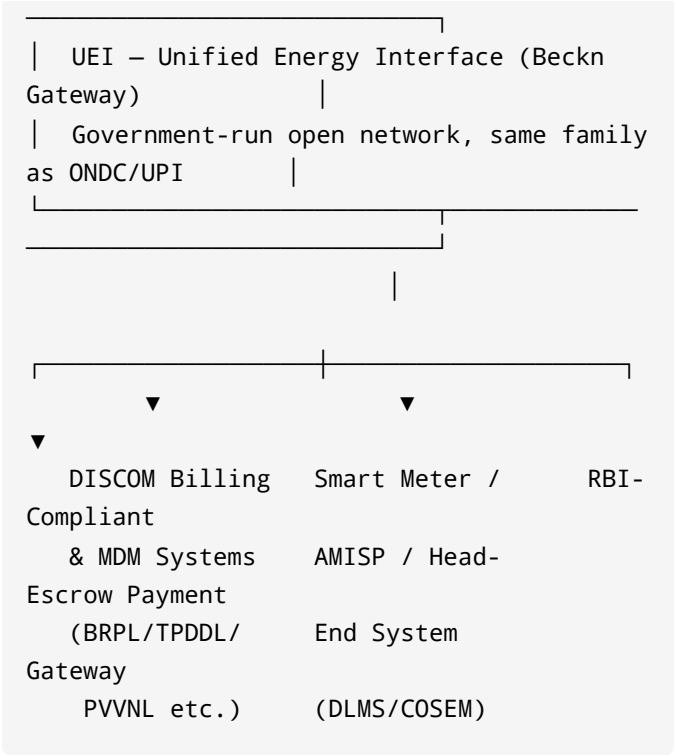
Version: 1.0 | **Companion to:** PRD v1.0 | **Date:** August 2026

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1. Architectural Positioning

SolarShare is designed as a **thin application layer**, not a ground-up energy platform. It sits on top of infrastructure that already exists or is being actively rolled out by the Government of India:





This mirrors exactly how Pulse Energy's SUPEREV app already operates as an authorized Trade Platform Provider (TPP) on the IES — reusing OCPP-based session management and ML-based demand optimization originally built for EV charging, rather than building a parallel energy network. SolarShare follows the same integration pattern for solar P2P trading instead of EV charging.

2. Technology Stack

Layer	Hackathon/MVP Choice	Rationale
Frontend (mobile)	React Native / Flutter	Cross-platform, fast to prototype
Frontend (chat-based)	WhatsApp Business API	Mirrors the real IES pilot UX — no app download friction
Backend	Node.js / Express or Python / FastAPI	Fast iteration, large talent pool for a student team
Database	PostgreSQL (or Firebase/Supabase for MVP)	Relational integrity for trades/ledger; Firebase for zero-cost prototyping
Ledger/Trust layer	SHA-256 hashed transaction records (MVP) → Permissioned blockchain (Hyperledger	Public blockchains are unnecessary and costly for a closed, KYC'd

Layer	Hackathon/MVP Choice	Rationale
	Fabric) at pilot stage	network of verified DISCOM customers; a permissioned ledger matches the IES's own "blockchain-based trust and validation layer" design
Hosting	Firebase / AWS Free Tier / Vercel	₹0 at MVP scale
Matching engine	Rule-based (MVP) → ML-based demand/surplus forecasting (Phase 2)	Start simple, add intelligence once real usage data exists

3. Integration Requirements

3.1 UEI / Beckn Protocol Registration

To operate as a real Trade Platform Provider, SolarShare must register as a **Beckn Application Provider (BAP)** on the UEI network (built on the open-source, India-made Beckn Protocol — the same protocol family underlying ONDC). The Beckn Gateway acts as the central registry connecting BAPs (demand-side apps like SolarShare) to BPPs (Beckn Platform Providers — supply-side/DISCOM systems), using standard `search → select → init → confirm` API calls.

3.2 Smart Meter Data

- Meter data communication follows **DLMS/COSEM protocol**, the standard mandated for Indian smart meters.
- Meters must comply with **IS 16444** (Indian standard for AC static smart meters) and relevant IEC standards (62052-11, 62053-22/23) for accuracy.
- Under **CEA Smart Meter Regulations 2023**, all net-metered connections require AMI-enabled bidirectional meters (Class 1.0 for single-phase residential, Class 0.5S for three-phase) that record import and export in separate registers.
- SolarShare does not read meters directly; it consumes verified data via the DISCOM's Advanced Metering Infrastructure Service Provider (AMISP) / Head-End System, exposed

through the UEI layer — the same pattern the live pilot already uses.

3.3 DISCOM Billing Integration

Trade settlements are **not** separate cash transactions between neighbors — per the live pilot design, they are reflected as cumulative adjustments in the buyer's and seller's regular DISCOM bills. SolarShare's backend must reconcile trade records against DISCOM billing cycles rather than attempt independent metering or billing.

3.4 Payment Settlement

Per the IES design, settlement uses an **RBI-compliant payment gateway with escrow**, ensuring assured payment without either party needing to trust the other directly. At MVP stage, this can be simulated with a mock wallet; at pilot stage, integration with an existing UPI/NPCI-compliant payment aggregator is required.

4. Data Flow (End-to-End)

1. Solar panel generates power → net meter records export in real time
2. Smart meter pushes reading to DISCOM's Head-End System via DLMS/COSEM

3. SolarShare (as BAP) queries available surplus via UEI/Beckn `search` call
4. Buyer's app receives listings, sends `select` → `init` → `confirm` for a trade
5. Trade is hashed and logged on the trust/ledger layer
6. Settlement instruction sent to escrow payment gateway
7. DISCOM billing system applies the net adjustment to both parties' next bill

5. Security & Compliance Requirements

- **Identity verification:** every user must be a verified DISCOM customer with valid metering credentials (matching the IES "secure digital identity" model) — prevents spoofed listings.
- **Data protection:** consumption/generation data handled under India's **Digital Personal Data Protection (DPDP) Act, 2023**.
- **Tamper evidence:** every trade record hashed (SHA-256 minimum) and chained to the prior record, so retroactive edits are detectable.
- **Payment security:** no direct peer-to-peer cash handling; all settlement via escrow to prevent fraud/non-payment disputes.

6. Non-Functional Requirements

Requirement	Target
Matching latency	< 5 seconds for a listing-to-suggestion response
Uptime	99.5%+ (platform is a value-add layer; grid supply itself must never be interrupted)
Scalability	Architecture must support scaling from 1 society (~50–200 meters) to multi-society (~5,000+ meters) without redesign
Fallback behavior	If no local match exists, buyer transparently falls back to standard grid supply — no manual intervention required

7. Testing Strategy

- **Unit tests:** matching algorithm, hash generation, price calculation logic
- **Integration tests:** simulated smart meter feed → listing → match → mock settlement

- **Regulatory conformance checks:** transaction fee calculation against the pilot's ₹0.42/kWh reference structure (split buyer/seller) before any real-money pilot
- **Load testing:** simulate a full housing society (200+ units) trading simultaneously during peak solar hours

8. Third-Party Dependencies

Dependency	Purpose
UEI / Beckn Gateway	Network registration, discovery, transaction protocol
DISCOM AMISP / Head-End System	Verified meter data
RBI-licensed payment aggregator	Escrow-based settlement
WhatsApp Business API	Low-friction onboarding and notifications
REC Limited (nodal agency)	BAP/TPP authorization pathway